



UNIVERSIDAD DE LAS FUERZAS ARMADAS - ESPE

DATE:

15/06/2026

CLASS:

AWD - 30716

WORKSHOP:

Product Backlog Validation

Nº) NOMBRES Y APELLIDOS..... No/10 Git ☐ Doc ☐  

01) DYLAN ALEXANDER ALVARADO PALACIOS..... /10

URL: link

Features: -6 4/10

Features

Feature	Description	Actor(s)	User Stories	PB Items
F-01	Authentication & Role-Based Access Control — Login, session management, and role-restricted module access.	All Actors	US-01	PB-01, PB-02, PB-03
F-02	Patient Management — Full CRUD of patient records with ID, age, legal representative, and consultation reason validations.	Dentist	US-02, US-03	PB-04, PB-05, PB-06, PB-07, PB-08, PB-17
F-03	Patient Business Logic — Server-side calculations on patient data: pediatric category classification, legal representative check, birthday countdown, consultation time, urgency level, and treatment complexity.	Dentist	US-02, US-03	PB-21, PB-22, PB-24, PB-26, PB-27, PB-31
F-04	Payment Management — Full CRUD of payment records with type (Deposit / Final), method (Cash / Card / Transfer), amount and date validations.	Receptionist	US-04, US-05, US-08	PB-09, PB-10, PB-11, PB-12

F-05	Payment Business Logic — Derived payment status, senior discount eligibility, contact priority score, outstanding patient balance, and daily revenue summary.	Receptionist	US-05, US-08	PB-23, PB-25, PB-28, PB-29
F-06	Inventory Management — Full CRUD of dental supplies with quantity, unit cost, order date, and expiration date validations and automatic status assignment.	Administrator	US-06, US-07	PB-13, PB-14, PB-15, PB-16
F-07	Inventory Analytics & Alerts — Financial analysis (asset valuation, restock budget, expiration losses, capital risk) and critical stock alerts.	Administrator	US-07, US-09	PB-15, PB-30
F-08	Security & Reporting — Encrypted credential storage, statistical dashboard, patient/inventory search, and PDF/Excel export.	All Actors	—	PB-17, PB-18, PB-19, PB-20

7.2.2 Payments — CRUD

Tag	Method	URI	Description
[CRUD]	POST	/fabuladental/payments	Record a new patient payment (Deposit or Final)
[CRUD]	PUT	/fabuladental/payments/:paymentId	Update a recorded payment to correct errors or status
[CRUD]	DELETE	/fabuladental/payments	Delete a payment record by ID (sent in request body)

The GET method is not documented

5.1 Implemented Functional Requirements

- RF-01: Complete CRUD of payments with types (Deposit/Final) and methods (Cash/Transfer/Card) — TypeScript + Prisma
- RF-02: Automatic derivation of payment status (Deposit → Partial, Final → Completed) in Business Logic layer
- RF-03: Complete CRUD of supplies with automatic expiration status calculation — TypeScript + Prisma
- RF-04: Automatic alerts for products with an expiration date ≤ 30 days (NextExpiration status)
- RF-05: Public treatment page accessible without authentication
- RF-06: RESTful APIs for each module (GET, POST, PUT, DELETE) — TypeScript/Express
- RF-07: Business Logic endpoints for financial analysis: asset valuation, restock provisioning, expiration losses, capital risk
- RF-08: Payment history endpoint with calculated status per payment record
- RF-09: Patient payment history endpoint filtered by patient ID
- RF-10: Legal representative validation endpoint for underage patients (PB-21)
- RF-11: Patient Business Logic endpoints — birthday countdown, senior discount, consultation time estimate, contact priority (PB-22–PB-25)
- RF-12: Patient urgency level assessment endpoint (BL Server) (PB-26)
- RF-13: Patient treatment complexity estimation endpoint (BL Server) (PB-27)
- RF-14: Outstanding patient balance summary endpoint per patient (BL Server) (PB-28)
- RF-15: Daily revenue summary endpoint grouped by payment method (BL Server) (PB-29)
- RF-16: Critical stock alert endpoint for inventory management (BL Server) (PB-30)
- RF-17: Pediatric category classification endpoint based on patient date of birth (BL Server) (PB-31)

5.2 Non-Functional Requirements

- RNF-01 Usability: Responsive interface with Bootstrap 5. Compatible with mobile and desktop.
- RNF-02 Performance: CRUD operations do not require full page reload (fetch API). BL communicates with CRUD via HTTP.
- RNF-03 Scalability: PostgreSQL (Supabase) with pgbouncer connection pooling for horizontal scaling.
- RNF-04 Portability: Deployable on GCP or any Node.js-compatible environment.
- RNF-05 Maintainability: Clean separation between CRUD and Business Logic servers. TypeScript strict mode enforced.
- RNF-06 Validation: Backend input validation — 10-digit IDs, positive amounts, non-future dates, valid enums, inventory business rules.
- RNF-07 Type Safety: All backend code compiled with TypeScript strict mode with zero type errors.

URI not implemented: /patients/urgency-level

1.6 Features Overview

treatment

The table below groups all Product Backlog items into the eight high-level features of the system. Each feature is owned by one or more actors and cross-references the corresponding User Stories and PB items.

Feature	Description	Actor(s)	User Stories	PB Items
F-01	Authentication & Role-Based Access Control — Login, session management, and role-restricted module access.	All Actors	US-01	PB-01, PB-02, PB-03
F-02	Patient Management — Full CRUD of patient records with ID, age, legal representative, and consultation reason validations.	Dentist	US-02, US-03	PB-04, PB-05, PB-06, PB-07, PB-08, PB-17
F-03	Patient Business Logic — Server-side calculations on patient data: pediatric category classification, legal representative check, birthday countdown, consultation time, urgency level, and treatment complexity.	Dentist	US-02, US-03	PB-21, PB-22, PB-24, PB-26, PB-27, PB-31
F-04	Payment Management — Full CRUD of payment records with type (Deposit / Final), method (Cash / Card / Transfer), amount and date validations.	Receptionist	US-04, US-05, US-08	PB-09, PB-10, PB-11, PB-12

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URI not implemented: /patients/treatment-complexity

Fabula Dental — Product Backlog v3.0 | Team: CodeExplorers | June 12, 2020

F-05	Payment Business Logic — Derived payment status, senior discount eligibility, contact priority score, outstanding patient balance , and daily revenue summary.	Receptionist	US-05, US-08	PB-23, PB-25, PB-28, PB-29
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URI not implemented: /payments/patients/:id/balance

URI not implemented: /payments/daily-revenue

F-07	Inventory Analytics & Alerts — Financial analysis (asset valuation, restock budget, expiration losses, capital risk) and critical stock alerts.	Administrator	US-07, US-09	PB-15, PB-30
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URI not implemented: /supplies/critical-stock

02) NELSON DAVID ANDINO MARTINEZ..... /10

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03) JILMAR JHOEL CALDERON CALVA..... /10

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04) ANDRES SEBASTIAN CARDENAS ERAZO..... /10

Validation of team 3:

Features 7/10

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2.2	Specialist Profiles	Desc: Creation of therapist accounts linked to specific departments. AC: Therapists have read/write access to their agenda and read-only cross-visibility to other departments.	5	High
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5.2	Cancellation Penalty Enforcement	Desc: Logic to penalize late cancellations. AC: Canceling within 24 hours triggers a non-refundable 100% penalty warning. The deposit is logged as earned revenue	5	High
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-1

6.2	Downloadable PDF Vouchers	Desc: Generate a digital receipt upon appointment confirmation. AC: PDF must include the Ser salud logo, Date, Time, Amount, Specialist, and a reminder to arrive 15 minutes early.	5	Med
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The next are right:

1.3	Dynamic Services Catalog	Desc: Display the clinical services (Traumatological, Neurological, Respiratory, Sports) directly on the landing page for quick access.	2	Med
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7.1 View All Therapies

GET <http://20.55.45.48:4000/api/terapias>

1. Add a transaction inspection inquiry structural node point path called **Therapies - Get All**.
2. Maintain the selection state indicator configuration option parameter fields exactly as standard **GET** context.
3. Supply the endpoint destination reference location address link mapping string: **http://20.55.45.48:4000/api/terapias**.
4. No payload bodies or special token credentials are required. Click **Send** to verify the return array formatting matches collection design guidelines.

7.2 View Therapies by Specialty

GET <http://20.55.45.48:4000/api/terapias?especialidad=Fisioterapia>

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
2.1	Frictionless Patient Login	Desc: Registration asks only for DNI and Full Name to prevent user drop-off. AC: System auto-generates a username (e.g., initial + surname) and sets the DNI as the permanent password.	5	High

6.2 Register User (Patient)

POST <http://20.55.45.48:4000/api/auth/register>

Applies strict validation of the Ecuadorian identification algorithm (Cédula).

Other Dependencies				
2.3	Admin Privilege Management	Desc: Administrator dashboard to manage clinic staff and services. AC: Admin can add new therapy areas and assign specialists dynamically.	3	Med

10.1 Get All Users

GET <http://20.55.45.48:4000/api/users>

Authorization: Bearer {{token_admin}}

1. Setup a collection tracking testing node and label it **Users - Get All (Admin)**.

3.1	Deferred Medical History	Desc: A non-mandatory clinical history form presented as a gentle prompt before the first booking. AC: Can be skipped by the patient and filled out later by the therapist during the physical session.	8	High
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5.2 Find Many Records (findMany)

POST <http://3.135.164.193:4000/db/query>

Extracts collection records mapped structurally to domain parameters.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
4.1	24-Hour Chronological Validation	Desc: Enforce a strict buffer for all non-emergency appointments. AC: The interactive calendar disables any time slots that are less than 24 hours away from the server's current timestamp.	8	High

9.1 Get Available Time Slots

GET <http://20.55.45.48:4000/api/citas/horarios-disponibles?medicoId=9&fecha=2026-06-06>
Authorization: Bearer {{token_jwt}}

1. Construct a query monitoring object and label it `Appointments - Get Available Slots`.

4.2	Cross-Department Anti-Overlap	Desc: Prevents a single patient from holding concurrent appointments. AC: If Patient A has Traumatology at 10:00 AM, the system blocks them from booking Neurology at 10:00 AM on the same day.	8	High
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9.1 Get Available Time Slots

GET <http://20.55.45.48:4000/api/citas/horarios-disponibles?medicoId=9&fecha=2026-06-06>
Authorization: Bearer {{token_jwt}}

1. Construct a query monitoring object and label it `Appointments - Get Available Slots`.

4.3	Internal Patient Referral	Desc: Allows a specialist to schedule a subsequent appointment for their patient. AC: Therapists can book slots in their own calendar or refer the patient directly into an available slot in another department.	5	Med
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9.2 Create Appointment

POST <http://20.55.45.48:4000/api/citas>

Authorization: Bearer {{token_jwt}}

1. Deploy a resource transaction reservation workflow sequence item module and label it **Appointments - Book Appointment**.

05) KERLY ANDREINA CHUQUI AGUINDA..... /10

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06) CAMERON VICTORIA DIAZ CAIZALUISA..... /10

URL: link

07) LENIN ISAAC ERAZO CAMPOS..... /10

08) CÉSAR LUIS GALARZA VILLALVA..... /10

09) BRAYAN PATRICIO GUALOTUÑA AMAGUAYA... /10

Features: (input) **9.0 / 10** (*1 issue*)

Conflicting URI: In the DELETE `/appointments/{appointment_id}` section, the example parameter points to a completely different route: `/appointment-services/...`

DELETE `/appointments/{appointment_id}`

Deletes an appointment from the database.

Parameters

Parameter	Location	Tipo	Required	Description
appointment_id	/appointment-services/9785f22b-3a89-4945-82a7-77d379756a78	string	Sí	Unique appointment identifier a eliminar

Codes de Respuesta

Code	Description
204	Appointment deleted successfully (no content)
422	Validation error

FEATURE: AUTHENTICATION & USERS	METHOD	URI

User Registration	POST	/auth/register
User Login	POST	/auth/login
Role Management (RBAC)	GET	/sessions/current
Authorization Middleware	GET	/sessions/current

FEATURE: BARBERSHOP MANAGEMENT	METHOD	URI
Create Barbershop	POST	/barbershops
Edit Barbershop	PUT	/barbershops/{shop_id}
View Barbershop Info	GET	/barbershops/{shop_id}
Edit Restriction	PUT	/barbershops/{shop_id}

FEATURE: BARBER MANAGEMENT	METHOD	URI
List Barbers	GET	/barbershops/{shop_id}/members
Manage Barbers	POST	/api/owner/barbers

Assign Barbers to Shop	POST	/barbershops/{shop_id}/members
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FEATURE: SERVICES	METHOD	URI
Create Service	POST	/barbershops/{shop_id}/services
Edit Service	PUT	/barbershops/{shop_id}/services/{service_id}
Delete Service	DELETE	/barbershops/{shop_id}/services/{service_id}
List Own Services	GET	/barbershops/{shop_id}/services
Public Service Ownership	GET	/api/customer/services
List Services Restriction	GET	/barbershops/{shop_id}/services

FEATURE: PRODUCTS	METHOD	URI
Create Product	POST	/barbershops/{shop_id}/products
Edit Product	PUT	/barbershops/{shop_id}/products/{product_id}

Delete Product	DELETE	/barbershops/{shop_id}/products/{product_id}
List Products	GET	/barbershops/{shop_id}/products

FEATURE: BOOKINGS / APPOINTMENTS	METHOD	URI
Create Appointment	POST	/appointments
View Availability	GET	/api/customer/available-times
Select Service	POST	/appointments/{appointment_id}/services
Barber Calendar	GET	/api/barber/appointments
Owner Dashboard	GET	/api/owner/appointments
Update Appointment Status	PATCH	/api/owner/appointments/{appointment_id}/status
Availability Validation	POST	/appointments

10) GABRIEL ANTHONY MOLINA GALLEGOS..... /10

URL:

<https://docs.google.com/document/d/1LnYWdqRkRqRcqdoAG9EvP2BPWPOGm-1lGNNqDdVosKc/edit?usp=sharing>

FTNumeral	URI	Tipo de Estado / Observación
FT-01	POST /users/register	Mapeado. Implementa el registro con RUC y autenticación segura.
	POST /users/login	
	POST /users/login/google	
	POST /users/complete-profile	

11) RICARDO ANDRES MONGE MIÑO..... /10

URL:

<https://docs.google.com/document/d/1LnYWdqRkRqRcqdoAG9EvP2BPWPOGm-1lGNNqDdVosKc/edit?usp=sharing>

FTNumeral	URI	Tipo de Estado / Observación
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	POST /users/login	
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12) ALEJANDRO XAVIER OBANDO ZAPATA..... /10

URL:

<https://docs.google.com/document/d/1LnYWdqRkRqRcqdoAG9EvP2BPWPOGm-1lGNNqDdVosKc/edit?usp=sharing>

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FTNumeral	URI	Tipo de Estado / Observación
FT-01	POST /users/register	Mapeado. Implementa el registro con RUC y autenticación segura.
	POST /users/login	
	POST /users/login/google	
	POST /users/complete-profile	
	POST /users/reset-password	

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13) ESTEBAN ARIEL QUIROGA BENITEZ..... /10

14) DAVID JOSUE RODRIGUEZ VILLARROEL..... /10

15) ANGEL JEANPIERRE SABANDO MORA..... /10

URL:<https://docs.google.com/document/d/1OGXhIXM2BEMKpCJBloOn-quwKto58-69/edit?usp=sharing&oid=109487110320888441043&rtpof=true&sd=true>

Accounting System – Product Backlog & Features Document					
Product Backlog Decomposition into Features by Module					
1. Features Summary Table					
ID	Feature Name	Primary Module	Sprint(s)	Lead(s)	Effort
FT-01	User Identity & Secure Access	Mod 1	1	D.R / E.Q	7 hrs
FT-02	Workspace & Project Isolation	Mod 1	1	A.S	7 hrs
FT-03	SRI Service Handshake	Mod 2	2-3	E.Q	7 hrs

FT-02 – Workspace & Project Isolation	
Field	Detail
Description	Creation of isolated local directories based on specific taxpayer RUCs and periods.
Objective	Prevent cross-contamination between different fiscal periods.

Module	
Module	Module 1 – Configuration & Setup
Associated RFs	RF-02, RF-03
Responsible	Angel Sabando
Business Rules	RN-01 (Tax periods)

Technical Tasks					
Task ID	Task	Rel. with	Responsible	Sprint	Hrs

3.1 Module 1: Authentication and Base Profile (Taxpayer)

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Method	Endpoint / URI	Description / Payload (JSON)
POST	/auth/register	Initial registration on the platform. Body: { email, password, firstName, lastName }
POST	/auth/login	Web app login. Validates against Supabase Auth. Returns: JWT Token + User data.
GET	/taxpayer/profile	Retrieves the authenticated taxpayer's profile information. Header: Authorization: Bearer <token>
PUT	/taxpayer/profile	Updates the taxpayer's profile information. Body: { firstName, lastName, phone }

16) CARLOS ALEXANDER TORRES PINCAY..... /10

URL: link

17) EVELYN HAYDEE VILLARREAL SALGADO..... /10

URL: link